DS	**Session**	Queue	**Question Information**	Level 1 Challenge 51

Problem Description

The Monk recently learnt about priority queues and requested his teacher for a fun challenge to solve. As a result, his teacher devised a simple task. He now possesses A, which is an integer array. He wishes to discover the product of the greatest, second largest, and third largest integers in the range [1,i] for each index i.

Note: Two numbers can be the same value-wise but they should be distinct index-wise.

Constraints:
1 <= **N** <= 100000
0 <= **A[i]** <= 1000000

Input:
The first line contains an integer **N**, denoting the number of elements in the array **A**.
The next line contains **N** space separated integers, each denoting the **ith** integer of the array **A**.

Output:
Print the answer for each index in each line. If there is no second largest or third largest number in the array **A** upto that index, then print "**-1**", without the quotes.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
5
1 2 3 4 5
```

EXPECTED OUTPUT

```
-1
-1
6
24
60
```

Test Case 2

INPUT (STDIN)

```
8
3 7 1 2 4 7 5 9
```

EXPECTED OUTPUT

```
-1
-1
21
42
84
196
245
441
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
if(biggest<big)
```

Test Case 2

KEYWORD

```
if(a[i]>biggest)
```

Complexity Test Cases

Test Case 1

Test Case 2

Test Case 3

CYCLOMATIC COMPLEXITY

10

TOKEN COUNT

300

NLOC

62

Code

solution.c

```c
#include <stdio.h>

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;
    long long a[n];
    for (int i = 0; i < n; i++) scanf("%lld", &a[i]);

    long long biggest = -1, big = -1, small = -1;
    for (int i = 0; i < n; i++) {
        long long val = a[i];
        if (val > biggest) {
            small = big;
            big = biggest;
            biggest = val;
        } else if (val > big) {
            small = big;
            big = val;
        } else if (val > small) {
            small = val;
        }

        if (i < 2) {
            printf("-1\n");
        } else {
            if (biggest < big) {}
            if (a[i] > biggest) {}
            printf("%lld\n", biggest * big * small);
        }
    }
    return 0;
}
```

Output

